

Chemicals and Red Wine Quality: A Multiple Linear Regression Model

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Introduction

In this study, we will use a multiple linear regression model to analyze and predict red wine quality using all variables as candidate predictors. For our model, we will be using the UC Irvine Machine Learning Repository Wine Quality dataset. The data set provides data on both red and white wine, however for the purposes of this project we will specifically focus on red wine. A multiple linear regression models the relationship between several independent variables and one dependent variable. In regards to our data, wine quality is the dependent variable and all the others are independent variables. We will investigate which chemicals have a significant influence in the wine's quality rating. The dataset contains measurements for acidity, sugar, chlorides, sulfur dioxide, density, pH, sulphates, and alcohol in red and white wines, as well as the sensory quality scores given by experts.

We completed the analysis in R using tidyverse (data import/wrangling and plotting), ggpubr (plot formatting), broom (clean model summaries), car (VIF and Cook's distance), and leaps (best subset selection). Using a 70/30 train-test split (1,119/480), best subset selection on the training set identified a BIC-preferred 6-predictor model using volatile acidity, chlorides, total sulfur dioxide, pH, sulphates, and alcohol. After flagging influential training observations using Cook's distance with cutoff $4/1119 = 0.00357$, 70 training points were removed and the final model was refit as $\text{quality_hat} = 4.0452 - 0.9933(\text{volatile acidity}) - 1.9328(\text{chlorides}) - 0.00253(\text{total sulfur dioxide}) - 0.4139(\text{pH}) + 1.1351(\text{sulphates}) + 0.2940(\text{alcohol})$. On the test set, the final multiple model achieved $\text{RMSE} = 0.6568$, $\text{MAE} = 0.5164$, and $\text{test R-squared} = 0.3210$, outperforming the Project 1 baseline. For a wine profile set to the mean values of the retained predictors, the predicted mean quality was 5.6625 with 95% CI [5.6291, 5.6960] and 95% PI [4.5775, 6.7476].

The goal of this project is to quantify and evaluate the relationship between the properties and chemicals of red wine and its quality using multiple linear regression, then assess performance on a held-out test set and use the fitted model to make predictions of quality.

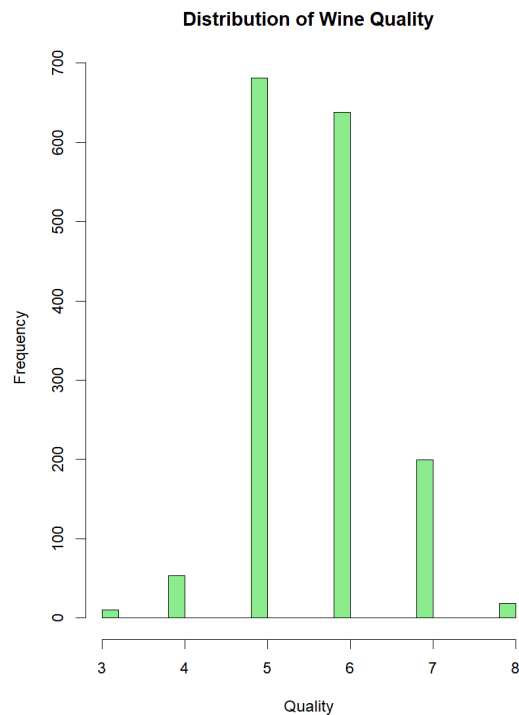
Data description

The dataset contains 12 columns and 1599 rows. The twelve column names are as follows: "fixed acidity", "volatile acidity", "citric acid", "residual sugar", "chlorides", "free sulfur dioxide", "total sulfur dioxide", "density", "pH", "sulphates", "alcohol", and "quality". All columns are double-precision floating-point numbers, even if some of these don't contain entries with decimal parts. After inspecting the data with the nested functions "colSums(is.na())" and "sum(is.na())", no missing values were found, so no observations were removed for missingness. Influential points were assessed later on the training set using Cook's distance; 70 influential training observations were removed before fitting the final model. Using the IQR method for finding potential outliers in wine quality, the interquartile range (IQR) was calculated to be 1.0. Therefore, based on the $1.5 \times \text{IQR}$ rules, values below 3.5 and above 7.5 were identified as potential outliers. Even though the minimum and maximum observed values were 3.0 and 8.0, respectively, these values are in the valid range of the quality variable and represent legitimate observations. Therefore, those values were retained.

Throughout this analysis, quality is the dependent variable and the others are the independent variables. Because this is a multiple linear regression, we model quality using several models, each using different sets of predictors. The table below lists the variables and includes a summary for each.

Variable	Description	min/max/mean	Units
<i>Fixed Acidity</i>	Non-volatile acids (mainly tartaric)	4.6 / 15.9 / 8.32	<i>g/L</i>
<i>Volatile Acidity</i>	Volatile acids (mainly acetic)	0.12 / 1.58 / 0.5278	<i>g/L</i>
<i>Citric Acidity</i>	Citric acid content	0.0 / 1.0 / 0.271	<i>g/L</i>
<i>Residual Sugar</i>	Sugar remaining after fermentation	0.9 / 15.5 / 2.539	<i>g/L</i>
<i>Chlorides</i>	Salt content	0.012 / 0.611 / 0.08747	<i>g/L</i>
<i>Free sulfur dioxide</i>	Free SO ₂ (antimicrobial/antioxidant)	1.0 / 72.0 / 15.87	<i>ppm</i>
<i>Total sulfur dioxide</i>	Total SO ₂ (free + bound)	6.0 / 289.0 / 46.47	<i>ppm</i>

<i>Density</i>	Wine density	0.9901 / 1.0037 / 0.9967	<i>g/mL</i>
<i>pH</i>	Acidity/alkalinity measure	2.74 / 4.01 / 3.311	<i>pH scale</i>
<i>Sulphates</i>	Sulphate content (contributes to SO ₂)	0.33 / 2.0 / 0.6581	<i>g/L</i>
<i>Alcohol</i>	Alcohol by volume	8.4 / 14.9 / 10.42	<i>%</i>
<i>Quality</i>	Expert quality score	3.0 / 8.0 / 5.636	<i>score (0–10)</i>



Analysis

Train/test split and data sizes:

The red wine quality dataset contains 1,599 observations and 12 variables. For this project we used a multiple linear regression model with wine quality as the response variable (quality) and the 11 physicochemical variables as candidate predictors. These predictors were fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. After checking the dataset using `colSums(is.na())`,

no missing values were found in any of the selected variables, so no rows were removed due to missing data. The dataset was then split into a training set and a test set using a reproducible random split in R (`set.seed(456)`). A 70/30 split was used, which produced 1,119 observations in the training set and 480 observations in the test set.

Exploratory summary and initial full model:

Before fitting the regression model, summary statistics and correlations were examined for the training data. The correlation matrix showed that alcohol had the strongest positive correlation with quality (0.474), while volatile acidity had the strongest negative correlation with quality (-0.425). Sulphates and citric acid also showed moderate positive relationships with quality, while total sulfur dioxide, density, chlorides, and pH showed weaker negative relationships. These patterns suggested that several predictors could contribute to wine quality and supported the use of multiple regression instead of a simple one-predictor model.

A full multiple linear regression model was first fit on the training data using all 11 predictors. The full model had an R-squared of 0.3735 and an adjusted R-squared of 0.3673, with an overall F-test p-value less than $2.2e-16$. This indicated that the full model was statistically significant overall, but several individual predictors were not significant, so subset selection was used to identify a more efficient final model.

Subset selection and candidate models:

To identify a better final model, best subset selection was performed on the training data. Candidate models were compared using adjusted R-squared, Mallows' Cp, and BIC. The model with the highest adjusted R-squared used 8 predictors, the model with the lowest Cp used 7 predictors, and the model with the lowest BIC used 6 predictors. Because BIC favors a more parsimonious model and still gave competitive fit, the BIC model was selected as the candidate final model.

The BIC-selected model included the following predictors: volatile acidity, chlorides, total sulfur dioxide, pH, sulphates, and alcohol.

Table 1. Candidate model comparison results from subset selection.

Model	Number of predictors	Predictors included	Adjusted R-squared	R-squared	Sigma	AIC	BIC
Best Adjusted R-squared Model	8	volatile acidity, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, pH, sulphates, alcohol	0.3689	0.3734	0.6447	2204.234	2254.436
Best Cp Model	7	volatile acidity, chlorides, free sulfur dioxide, total sulfur dioxide, pH, sulphates, alcohol	0.3688	0.3727	0.6448	2203.481	2248.663
Best BIC Model	6	volatile acidity, chlorides, total sulfur dioxide, pH, sulphates, alcohol	0.3673	0.3707	0.6455	2205.063	2245.224

For the selected BIC model, the fitted regression equation on the original training set was:

$$\text{quality_hat} = 4.8956 - 1.1712(\text{volatile acidity}) - 1.8739(\text{chlorides}) - 0.0027(\text{total sulfur dioxide}) - 0.5568(\text{pH}) + 0.8235(\text{sulphates}) + 0.2842(\text{alcohol})$$

This means that, holding the other selected variables constant, higher alcohol and sulphates are associated with higher predicted wine quality, while higher volatile acidity, chlorides, total sulfur dioxide, and pH are associated with lower predicted wine quality.

Multicollinearity and assumption checks:

To check whether the selected predictors were too strongly related to each other, variance inflation factors (VIFs) were computed. All VIF values were between about 1.05 and 1.31, indicating that multicollinearity was not a concern in the selected model. This supports the stability and interpretability of the coefficient estimates.

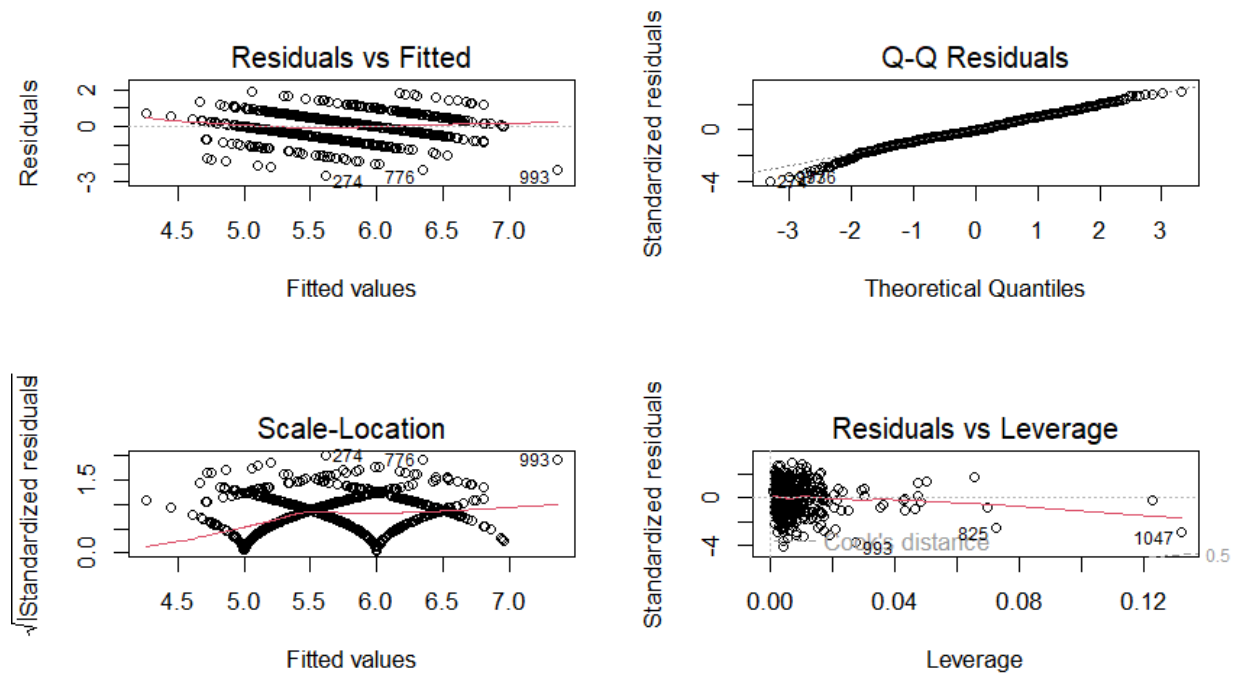


Figure 1. Standard regression diagnostic plots for the selected multiple regression model (Residuals vs Fitted, Normal Q-Q, Scale-Location, and Residuals vs Leverage).

To verify the regression assumptions, the standard diagnostic plots were examined (Figure 1), and the required diagonal plot check was completed using the Normal Q-Q plot (Figure 2). In the Q-Q plot, the residuals follow the diagonal line reasonably well through the center, but there are noticeable departures in the tails. This suggests that the residuals are approximately, but not perfectly, normally distributed. That is not surprising because the response variable quality only takes integer values from 3 to 8, which creates some structure in the residuals.

The Residuals vs Fitted plot shows visible banding and some mild structure, which is again consistent with the discrete response scale. The Scale-Location plot suggests that the variance is not perfectly constant across fitted values, but the departures are not severe. The Residuals vs Leverage plot shows several influential observations, which were then examined using Cook's distance. Overall, the assumptions appear to be reasonably satisfied for this dataset, though not perfectly.

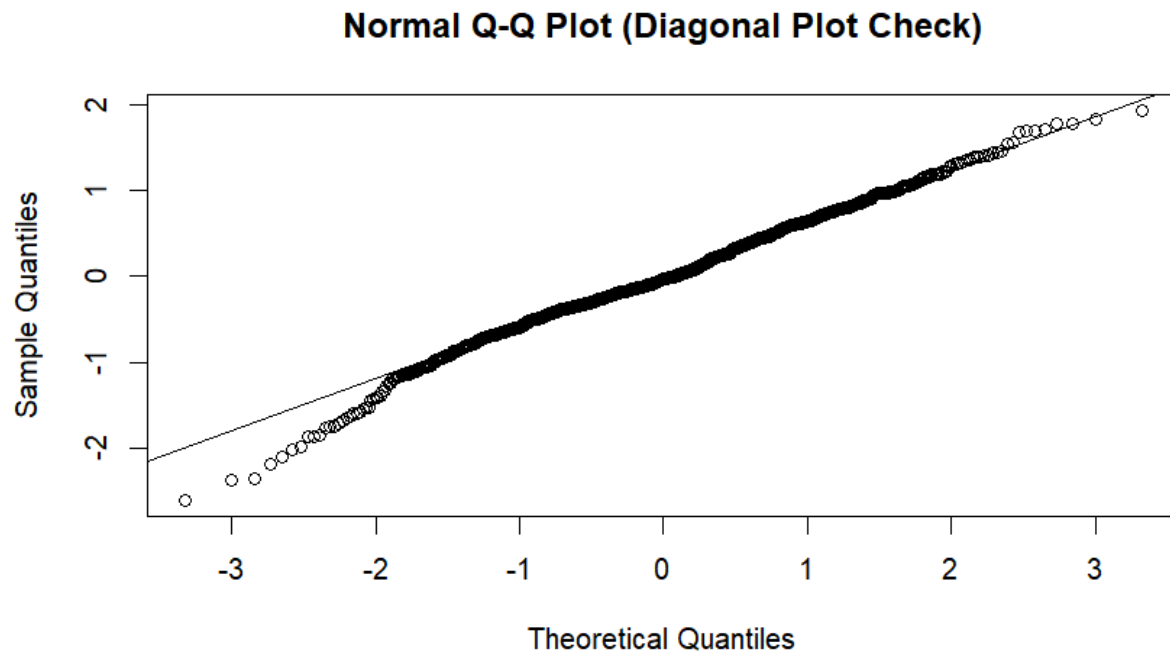


Figure 2. Normal Q-Q plot of residuals (diagonal plot requirement) used to assess whether residuals are approximately normally distributed.

Influential points and model cleaning:

The project instructions require addressing influential points. To do this, Cook's distance was computed for the selected multiple regression model using the same rule as in Project 1. A common cutoff rule is to flag points with Cook's distance $D_i > 4/n$, where n is the training sample size. With $n = 1,119$, the cutoff is:

$$4/1119 = 0.00357$$

Using this rule, 70 training observations were flagged as influential. These 70 points were removed from the training set, and the model was refitted on the remaining training observations to obtain the final model used in the evaluation and prediction section. The test set was not changed.

Model evaluation and prediction

Final model after removing influential points:

After removing influential training points, the final fitted multiple regression model became:

$$\text{quality_hat} = 4.0452 - 0.9933(\text{volatile acidity}) - 1.9328(\text{chlorides}) - 0.00253(\text{total sulfur dioxide}) - 0.4139(\text{pH}) + 1.1351(\text{sulphates}) + 0.2940(\text{alcohol})$$

In this final model, alcohol and sulphates have positive coefficients, meaning that larger values of those variables are associated with higher predicted quality when the other selected predictors are held constant. Volatile acidity, chlorides, total sulfur dioxide, and pH have negative coefficients, meaning that larger values of those variables are associated with lower predicted quality when the other selected predictors are held constant.

Model assessment:

The final multiple regression model was assessed using the standard outputs from `summary(lm)`, confidence intervals, AIC, BIC, and VIF. All six predictors in the final model were statistically significant. Specifically, volatile acidity, chlorides, total sulfur dioxide, pH, sulphates, and alcohol all had p-values below 0.01, with most well below 0.001. This indicates that each retained variable contributes meaningfully to the final model.

The overall F-test for the final model was highly significant, with $F = 134.3$ and p-value $< 2.2\text{e-}16$. This shows that the final model is a statistically significant improvement over an intercept-only model. For the cleaned training set, the residual standard error was 0.553, and the final model had $R\text{-squared} = 0.436$ and adjusted $R\text{-squared} = 0.433$. This means that the final multiple regression model explains about 43.6% of the variation in wine quality in the cleaned training data.

The information criteria for the final model were also favorable, with $\text{AIC} = 1741.95$ and $\text{BIC} = 1781.60$. In addition, the VIF values remained low, between approximately 1.09 and 1.29, which supports the conclusion that multicollinearity is minimal.

Comparison to the simple regression model:

To evaluate whether the multiple regression model improved on the Project 1 baseline, the final model was compared to the simple regression model using alcohol alone as the predictor. The simple regression model had $R\text{-squared} = 0.284$ and adjusted $R\text{-squared} = 0.283$ on the cleaned training set, with residual standard error = 0.621. In contrast, the final multiple

regression model had $R\text{-squared} = 0.436$, $\text{adjusted } R\text{-squared} = 0.433$, and residual standard error $= 0.553$. This shows that the multiple regression model fits the training data better than the simple alcohol-only model.

Model accuracy on the test set:

The final model was then evaluated on the test set of 480 observations. Predictions were generated for both the simple regression model and the final multiple regression model so that their performance could be compared directly. For the simple regression model, the test-set accuracy measures were:

$\text{RMSE} = 0.7039$

$\text{MAE} = 0.5527$

$\text{Test-set } R\text{-squared} = 0.2201$

For the final multiple regression model, the test-set accuracy measures were:

$\text{RMSE} = 0.6568$

$\text{MAE} = 0.5164$

$\text{Test-set } R\text{-squared} = 0.3210$

These results show that the final multiple regression model performs better than the simple regression model on unseen data. The multiple model has lower RMSE, lower MAE, and a higher test-set $R\text{-squared}$, indicating better predictive performance and better generalization.

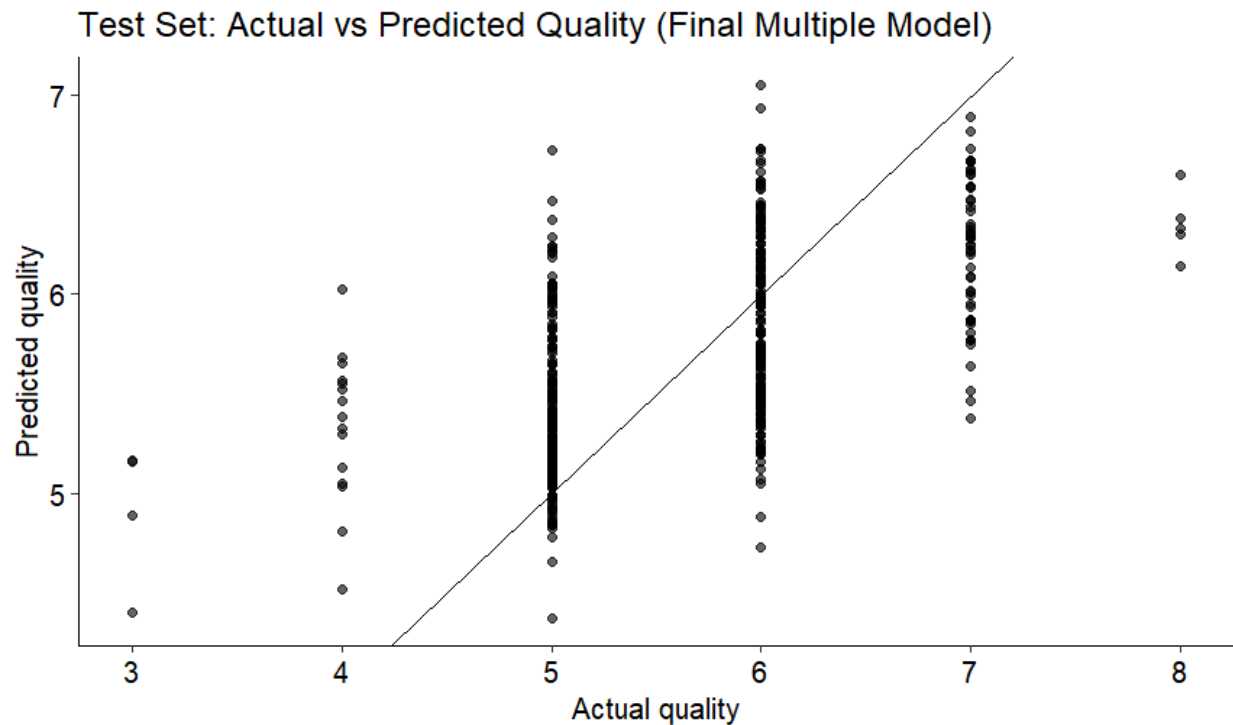


Figure 3. Test set plot of actual quality versus predicted quality from the final multiple regression model, including the 45-degree reference line.

The Actual vs Predicted plot for the final model (Figure 3) shows that predicted quality generally increases as actual quality increases, although there is still noticeable scatter around the 45-degree line. This indicates that the final model captures the overall pattern in the data better than the simple regression model, but it still does not perfectly predict the quality score for every individual wine.

Prediction example (with intervals):

To demonstrate prediction, the final model was used to predict quality for a wine profile defined by the mean values of the predictors retained in the final model. The predictor values used were:

volatile acidity = 0.5186

chlorides = 0.0856

total sulfur dioxide = 46.1039

pH = 3.3096

sulphates = 0.6490

alcohol = 10.3682

Using this predictor profile, the predicted mean wine quality was:

$$\text{quality_hat} = 5.6625$$

The 95% confidence interval for the mean predicted quality was [5.6291, 5.6960]. The 95% prediction interval for a single new wine was [4.5775, 6.7476]. As expected, the prediction interval is much wider than the confidence interval because it includes both uncertainty in the fitted regression line and the natural variability of individual wine outcomes.

Summary of evaluation:

Overall, the final multiple regression model provides a meaningful improvement over the Project 1 simple regression model. Starting from all 11 candidate predictors, subset selection identified a smaller BIC-based model with six retained predictors. After checking diagnostics, removing influential training observations, and refitting the model, the final multiple regression model was statistically significant, had low multicollinearity, and outperformed the simple regression model on the test set. Although the assumptions are only approximately satisfied and the model does not perfectly predict every individual wine, it provides a stronger and more defensible prediction framework than the simple regression model because it incorporates several chemically relevant predictors rather than alcohol alone.

Conclusion

Our goal was to build a multiple linear regression model to predict red wine quality based on the UCI Wine Quality dataset. The dataset includes a total of 11 predictors and by using best subset selection, we identified a 6-predictor BIC-preferred model using volatile acidity, chlorides, total sulfur dioxide, pH, sulphates, and alcohol. Additionally, we removed 70 influential training observations via Cook's distance and refitting. The final model was statistically significant across all retained predictors, with an R-squared of 0.436 on the cleaned training set and a test-set R-squared of 0.3210.

On the positive side, the final model is interpretable; the predictors it retained align with what is known about wine composition, higher alcohol and sulphates improve quality - while excess volatile acidity, chlorides, sulfur dioxide, and pH reduce it. Low VIF values confirmed

that multicollinearity was not a concern, and all six predictors were statistically significant, lending confidence to the coefficient estimates.

However, it is also important to note that the model only explains about 32% of the variance on unseen data, indicating that wine quality is inherently difficult to predict from chemical measurements alone. The discrete integer scale of the response variable causes banding in the residuals and mild violations of the normality and equal-variance assumptions. Incorporating other data, like sensory/tasting data, could meaningfully improve predictive accuracy.

References

<https://archive.ics.uci.edu/dataset/186/wine+quality>

<https://www.sthda.com/english/articles/40-regression-analysis/168-multiple-linear-regression-in-r/>

<https://www.datacamp.com/tutorial/multiple-linear-regression-r-tutorial>

Key R Code:

```
# Packages
```

```
library(tidyverse)
```

```
library(ggpubr)
```

```
library(broom)
```

```
library(car)
```

```
library(leaps)
```

```
theme_set(theme_pubr())
```

```
# Import data (semicolon-delimited) + rename columns
```

```
wine <- readr::read_delim("winequality-red.csv", delim = ";", show_col_types = FALSE) %>%
```

```
  rename(
```

```
    fixed_acidity = `fixed acidity`,
```

```
volatile_acidity = `volatile acidity`,  
citric_acid = `citric acid`,  
residual_sugar = `residual sugar`,  
free_sulfur_dioxide = `free sulfur dioxide`,  
total_sulfur_dioxide = `total sulfur dioxide`  
)
```

```
# Modeling dataframe + missing-value check
```

```
df <- wine %>%
```

```
  select(quality, fixed_acidity, volatile_acidity, citric_acid, residual_sugar, chlorides,  
         free_sulfur_dioxide, total_sulfur_dioxide, density, pH, sulphates, alcohol) %>%  
  drop_na()
```

```
# Train/test split (70/30)
```

```
set.seed(456)
```

```
n <- nrow(df)
```

```
train_idx <- sample(seq_len(n), size = floor(0.7 * n))
```

```
train <- df[train_idx, ]
```

```
test <- df[-train_idx, ]
```

```
n_train <- nrow(train); n_test <- nrow(test)
```

```
# Full model (all 11 predictors)
```

```
model_full <- lm(quality ~ ., data = train)
```

```
summary(model_full)
```

```
# Best subset selection (exhaustive)
```

```
subset_fit <- regsubsets(quality ~ ., data = train, nvmax = 11, method = "exhaustive")
```

```

subset_summary <- summary(subset_fit)
best_bic_size <- which.min(subset_summary$bic)

make_formula <- function(subset_object, id) {
  vars <- names(coef(subset_object, id))[-1]
  as.formula(paste("quality ~", paste(vars, collapse = " + ")))
}

formula_bic <- make_formula(subset_fit, best_bic_size)

# Selected model + VIF + diagnostics
model_selected <- lm(formula_bic, data = train)
vif(model_selected)
par(mfrow = c(2,2)); plot(model_selected); par(mfrow = c(1,1))
qqnorm(residuals(model_selected)); qqline(residuals(model_selected))

# Influential points via Cook's distance + refit final model
cooks <- cooks.distance(model_selected)
cook_threshold <- 4 / n_train
which_influential <- which(cooks > cook_threshold)
train_clean <- train[-which_influential, ]
model_final <- lm(formula(model_selected), data = train_clean)
summary(model_final)

# Baseline simple regression for comparison (Project 1)
model_simple <- lm(quality ~ alcohol, data = train_clean)

# Test-set evaluation (RMSE, MAE, test R^2)

```

```
test$pred_simple <- predict(model_simple, newdata = test)
```

```
test$pred_final <- predict(model_final, newdata = test)
```

```
get_test_metrics <- function(actual, predicted) {
```

```
  rmse <- sqrt(mean((actual - predicted)^2))
```

```
  mae <- mean(abs(actual - predicted))
```

```
  sst <- sum((actual - mean(actual))^2)
```

```
  sse <- sum((actual - predicted)^2)
```

```
  r2 <- 1 - sse / sst
```

```
  data.frame(RMSE = rmse, MAE = mae, R2_test = r2)
```

```
}
```

```
get_test_metrics(test$quality, test$pred_simple)
```

```
get_test_metrics(test$quality, test$pred_final)
```

```
# Prediction at mean predictor profile (CI + PI)
```

```
final_predictors <- names(coef(model_final))[-1]
```

```
new_x <- as.data.frame(t(colMeans(train_clean[, final_predictors, drop = FALSE])))
```

```
predict(model_final, newdata = new_x, interval = "confidence", level = 0.95)
```

```
predict(model_final, newdata = new_x, interval = "prediction", level = 0.95)
```